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Accelerated high-solids anaerobic digestion of sewage sludge using low-temperature thermal pretreatment

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ABSTRACT

High-solids anaerobic digestion (HSAD) of sewage sludge is a promising alternative to conventional anaerobic digestion due to its use of smaller digesters and its reduced energy requirement; however, long digestion time is a key shortcoming. This study investigated whether low-temperature thermal pretreatment of sludge could accelerate high-solids digestion. The results showed that after 30 min treatment at 60, 70 and 80 °C, 9.1, 13.0 and 16.6% of sludge solids, respectively, were disintegrated and sludge flow indices increased from 0 to 0.098, 0.195 and 0.249, respectively. As a consequence, thermal pretreatment at 60, 70 and 80 °C increased the quantity of accessible substrates and decreased sludge viscosity such that, compared to the control, biogas production in the batch anaerobic digestion experiments increased by 7.3, 15.6 and 24.4%, respectively, and the apparent kinetic constants increased from 0.19 d^{-1} to 0.29, 0.39 and 0.39 d^{-1}, respectively. A high concentration of volatile fatty acids and a low pH resulted from thermal pretreatment, but neither of these exerted a significant negative effect on the subsequent HSAD. The optimal pretreatment condition was a temperature of 70 °C for 30 min. In the semi-continuous pilot scale HSAD experiment with optimal pretreatment, biogas yield increased by 11% compared to that with no pretreatment, and the digestion time decreased from 22 to 15 days. The improved digestion following low-temperature thermal pretreatment confirmed the efficacy of this technique to accelerate high-solids anaerobic digestion, as well as increase biogas yield.

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1. Introduction

Bioenergy recovery from organic wastes is developing rapidly around the world because it can avoid the environmental pollution resulting from untreated wastes and reduce reliance on fossil fuel (Ratanatamskul et al., 2014; Neumann et al., 2015). Sewage sludge, a mostly organic waste discharged from municipal wastewater treatment plants (WWTPs) as a byproduct, has tremendous potential for bioenergy recovery because thousands of WWTPs have been built worldwide during the last few decades. Anaerobic digestion is a widely used method to convert the organic matter in sludge to biogas (Nghiem et al., 2014). Although this process is a great success in most cases, its application is limited by spatial and economic factors. Conventional anaerobic digestion (CAD) commonly treats sludge with a total solids (TS) content of 2–5%, because the high viscosity of sludge at greater TS hampers mixing and mass transfer (Jolis, 2008). Due to the high water content in sludge, traditional digesters commonly require a huge operating volume; consequently, application of CAD is difficult in small WWTPs and in highly urbanized areas where space is limited (Duan et al., 2012). On the other hand, the organic content in sludge solids is sometimes lower than 50% due to special wastewater characteristics and treatment processes. During anaerobic digestion, the feedstock with low organic content usually results in low biogas production, and correspondingly reduces the benefit of anaerobic digestion.

To increase the applicability of anaerobic digestion, high-solids anaerobic digestion (HSAD) could be a feasible alternative to CAD. The solids concentration in HSAD reactors increased to 10–20%, and thus, smaller digesters requiring less energy for heating can be utilized (Guendouz et al., 2008). In addition, the volumetric biogas production (VBP) of high-solids digesters is significantly greater than that of conventional digesters (Liao et al., 2014). High-solids sludge can be readily obtained either by diluting dewatered sludge that has a TS content of approximately 20% or

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thermally treated sludge (Fig. 1). Higher treatment temperature apparently led to more production of VFAs, but the effect of treatment duration was limited. VFAs can also be produced due to the strengthened activity of thermophilic or hyper-thermophilic bacteria at 70 °C, but this effect was observed only after 9 h treatment (Ferrer et al., 2008). Although the produced VFAs increased the accessible substrate for methanogenesis, the accumulated undissociated VFAs possibly diffused passively into cells and disrupted homeostasis. The VFAs concentration of 400 mg/L was recognized to be the threshold for slight inhibition during HSAD (Duan et al., 2012). Therefore, the influence of the produced VFAs on the subsequent anaerobic digestion should be noted. The low pH of about 6.0 was not the best condition for anaerobic digestion. However, a feed sludge with low pH may improve the acid–base balance in HSAD, which can create a weak alkaline environment in digestate (Duan et al., 2012; Liao et al., 2014).

The increased soluble nitrogen in the thermally treated sludge demonstrated the degradation of proteins, but the deamination of amino acids originating from proteins was limited. Moreover, the effect of treatment temperature on the concentration of soluble ammonia was slight. During HSAD, there was no inhibition or only slight inhibition when TAN was lower than 2000 mg/L (Duan et al., 2012). Therefore, the dissolved ammonia during thermal pretreatment would not impact the subsequent anaerobic digestion. Due to the increase in soluble ammonia, the alkalinity of the thermally treated sludge increased a little, but the ratio of VFAs to ALK was still much higher than the threshold below which stable operation of anaerobic digestion would be threatened (Duan et al., 2012).

Thus, among the released substances during thermal treatment, the high concentration of VFAs was a key risk that could cause the subsequent anaerobic digestion to deteriorate. The effect of increased VFAs was further addressed in the batch anaerobic digestion experiments. The amino-carbonyl reaction (browning reaction) is another potential adverse factor arising during thermal pretreatment of sludge. The brown color of a sludge lysate has been attributed to the products of the amino-carbonyl reaction, and some of the products have been shown to be difficult to degrade. In the present study, the sludge supernatant did not turn brown at low temperatures, indicating that amino-carbonyl reactions did not occur.

3.2. Effect of thermal treatment on sludge viscosity

Sludge viscosity influences its fluid properties and thereby determines stirring performance. To ensure adequate mixing, high energy input for stirring is necessary in HSAD of sewage sludge. If the viscosity of high-solids sludge could be decreased by thermal pretreatment, the energy required for stirring would be lessened and the mass transfer would be improved. Based on the Ostwald (power law) model, the relationship between apparent viscosities and shear rates can be expressed as Eq. (4).

$$\mu = k \cdot \gamma^{n-1} \tag{4}$$

In Eq. (4), γ is the shear rate (s^{-1}), k is the flow consistency index ($Pa \cdot s^n$), μ is the apparent viscosity ($Pa \cdot s$) and n is the flow behavior index. To gather data for use in Eq. (4), the variation in sludge viscosities as a function of shear rates was measured twice at different treatment temperatures (Fig. 2), and the parameters describing the graphical relationships were also calculated (Table 1). The measured data were well described by the Ostwald model, indicating that the sludge behaved as pseudoplastic fluid.

The two viscosity tests showed that sludge flowability increased as treatment temperature increased, and the values of n had a good linear correlation with treatment temperatures (R^2 were 0.9775 and 0.9944 for the two tests). The relationship between treatment temperature and sludge flowability can be possibly explained by the status of individual sludge particles in the suspension. In high-solids sludge, sludge particles are linked or bound by extracellular polymeric substances (EPS) in their surface layer, leading to a large frictional resistance and thereby a high viscosity. During thermal treatment, some EPS, including carbohydrates, proteins and lipids, were detached, dissolved and even decomposed (Appels et al., 2010), which probably disrupted the links between sludge particles and lessened the apparent viscosity. A similar observation was reported by Farno et al. (2014).

The effect of thermal treatment on sludge viscosity has always been reported to be irreversible because the decomposition of proteins, carbohydrates and lipids is irreversible (Baudez et al., 2013; Farno et al., 2014). However, the viscosity of the thermally treated sludge in the present research indeed recovered in some degree when the sludge temperature declined from 50 to 80 °C during pretreatment to 33 °C during digestion. After the pretreatment at 80 °C, sludge viscosity decreased by 48.6% at a shear rate of 2.1 s^{-1}, compared to the viscosity of untreated sludge. When the sludge temperature returned to 33 °C, the sludge viscosity increased, but it was still 32.2% lower than that of the untreated sludge. Previous research proved that proteins and carbohydrates in the EPS network are only partly degraded during short-duration, low-temperature thermal treatment (Appels et al., 2010). Therefore, in the present study, as sludge temperature decreased the undecomposed EPS possibly intertwined again and also relinked to the surface of sludge particles, such that the sludge became sticky again. Nevertheless, due to the degradation of some EPS and the disintegration of flocs, the recovered viscosities were still lower than the viscosity of the untreated sludge.

In general, diffusive coefficients are inversely proportional to fluid viscosities, and their relationship can be expressed by Stocks–Einstein formula (Squires and Brady, 2005). Consequently, the decreased viscosity resulted from thermal pretreatment can enhance the mass transfer in digestate. On the other hand, the power requirement for stirring is positively related to viscosity, and thus, energy consumption by stirring decreases owning to thermal treatment.

3.3. Effect of thermal pretreatment on batch HSAD

In theory, thermal pretreatment of sludge can improve HSAD by enhancing the solubilization and hydrolysis of organic substances in the sludge solids. However, thermal pretreatment also can possibly cause the subsequent anaerobic digestion process to deteriorate because the treatment releases high concentration of VFAs and depresses pH. These two possibilities were addressed in batch anaerobic digestion experiments.

The negative effects did not occur and thermal pretreatment accelerated HSAD significantly (Fig. 3). After thermally treated sludge was mixed with the inoculum containing soluble VFAs of 266.5 ± 15.3 mg/L, the soluble VFAs in the thermally treated sludge decreased to 600–750 mg/L (Table 2). This concentration still possibly resulted in a slight inhibition of digestion (Duan et al., 2012), but the biogas production did not deteriorate in the present study because VFAs were rapidly converted to biogas in the initial stage of digestion. When the digestion was completed, the concentration of soluble VFAs decreased to below 200 mg/L. On the other hand, the pH also increased to about 7.6 after treated sludge was mixed with the inoculum, and then pH increased further to 8.1 (Table 2) due to the consumption of VFAs and the release of ammonia. Thus, the variations in VFA concentration and pH, resulting from thermal pretreatment, did not hamper the performance of the batch HSAD.

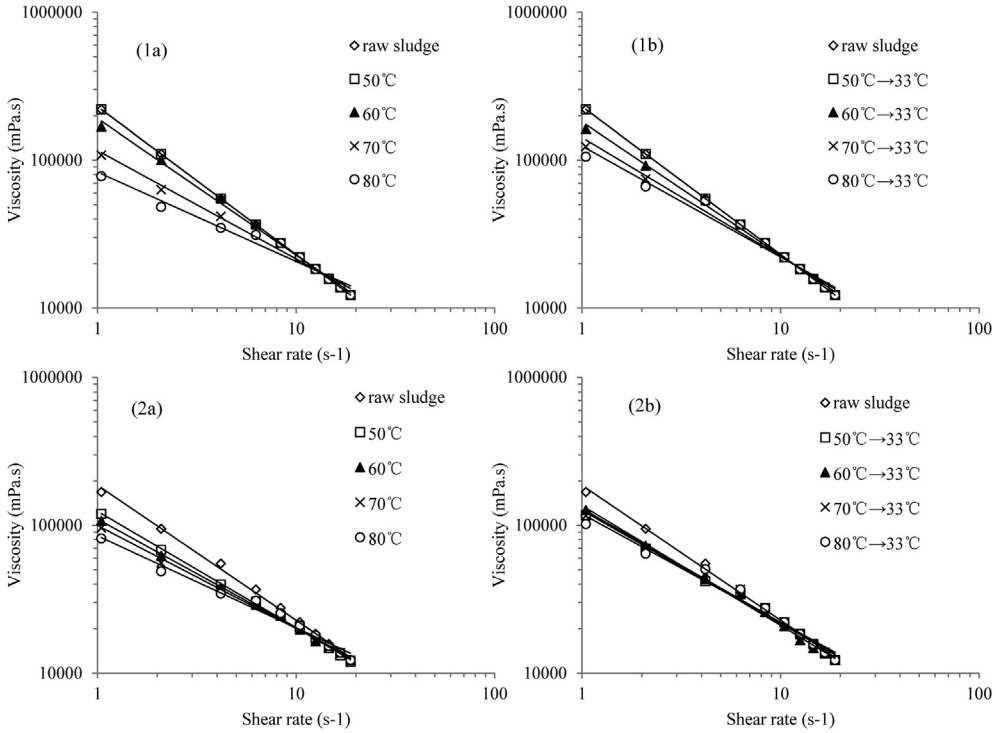

Fig. 2. Variation in sludge viscosity after 30 min thermal treatment at different temperatures ("1" and "2" denote the two tests; "a" indicates the variation after thermal treatment; "b" indicates the variation after the thermally treated sludge was cooled to 33 °C).

Table 1
Parameters of the power law model describing sludge viscosity after 30 min thermal treatment at different temperatures.

Temperature (°C)	k_1 (mPa·s^n)	n_1	R^2	k_2 (mPa·s)	n_2	R^2
25 (raw)	230,283	0	1.0000	186,787	0.086	0.9973
25 → 50	230,200	0	1.0000	124,410	0.213	0.9986
25 → 60	191,377	0.077	0.9972	110,183	0.262	0.9982
25 → 70	116,237	0.267	0.9886	99,671	0.303	0.9934
25 → 80	83,089	0.396	0.9680	84,913	0.377	0.9809
50 → 33	230,464	0	1.0000	126,386	0.234	0.9921
60 → 33	181,388	0.098	0.9965	135,900	0.190	0.9959
70 → 33	142,393	0.195	0.9856	129,757	0.230	0.9863
80 → 33	124,721	0.249	0.9730	119,405	0.266	0.9730

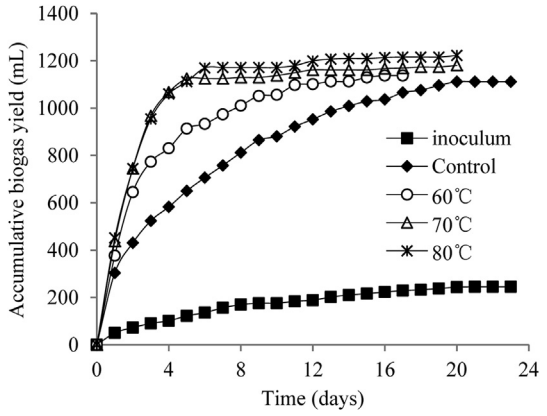

Fig. 3. Accumulative biogas yield of high-solids sludge (TS 15%) after 30 min thermal pretreatment at different temperatures (the "inoculum" is the batch experiment using only inoculum and the "control" is the batch experiment without thermal pretreatment).

In contrast to the reductions in SCOD and VFAs due to thermal treatment, the concentration of TN in the supernatant, mainly derived from ammonia, increased significantly during the digestion of the thermally treated sludge (Table 2). However, the concentrations of TAN and FAN were very close to the thresholds of TAN (2000 mg/L) and FAN (250 mg/L) required for anaerobic digestion, and possibly resulted in a slight inhibition to anaerobic processes (Duan et al., 2012). The release of ammonia also increased the alkalinity, which counteracted the effect of the high concentration of VFAs in the thermally treated sludge. A similarly weak alkaline environment was observed during continuous HSAD in a previous study (Liao and Li, 2015).

When the digestion completed in 20 days, the specific biogas production per unit VS added (SBP_{ad}) increased by 7.3%, 15.6% and 24.4%, respectively after 30 min thermal pretreatment at 60 °C, 70 °C and 80 °C (Table 3), compared to that from the control digesters. These increments were close to the previously reported value of 13.4% reported by Kim et al. (2003) after 30 min thermal treatment at 121 °C, and were only a little lower than the value of 25% obtained after thermal pretreatment at 190 °C (Bougrier et al., 2007). Compared with high-temperature thermal pretreatments reported elsewhere, the low-temperature thermal pretreatments in the present study were similarly effective in enhancing biogas production from anaerobic digestion. These results indicate that for anaerobic microorganisms, the accessibility of organic substances in sludge solids was improved due to low-temperature thermal pretreatment, even if these substances were not dissolved into the liquid phase to the same extent as achieved in high-temperature thermal pretreatment. On the other hand, the increase in SBP_{ad} was less than 30% even though the digestion time was long enough, suggesting that the increase in available biodegradable substances in feed sludge was limited. The SBP_{ad} values were lower than the common range of 300–500 mL/g associated with anaerobic digestion, which could be attributed to inadequate agitation using

Table 2
Characteristics of sludge supernatant before (BeD) and after (AfD) anaerobic digestion of thermally treated high-solids sludge.

Parameters		Control	60 °C	70 °C	80 °C
SCOD (mg/L)	BeD	2135.2 ± 85.9	7538.9 ± 75.4	12,003.8 ± 300.5	14,214.7 ± 106.6
	AfD	2324.2 ± 81.0	3898.3 ± 445.2	6531.5 ± 445.2	5157.6 ± 40.5
VFAs (mg/L)	BeD	198.6 ± 6.0	617.3 ± 8.9	629.2 ± 20.8	727.2 ± 11.9
	AfD	204.6 ± 16.8	163.0 ± 8.4	195.7 ± 12.6	130.3 ± 4.2
pH	BeD	7.6	7.7	7.8	7.6
	AfD	8.0	8.1	8.1	8.1
TN (mg/L)	BeD	822.6 ± 11.8	1555.9 ± 33.7	1677.7 ± 90.8	1960.1 ± 50.5
	AfD	1857.0 ± 33.3	2068.0 ± 28.5	2758.0 ± 23.8	2845.4 ± 156.9
TAN (mg/L)	BeD	771.0 ± 23.7	958.1 ± 32.7	1009.5 ± 13.3	1045.1 ± 4.1
	AfD	1803.3 ± 33.6	1951.8 ± 42.0	2112.2 ± 59.4	2040.9 ± 33.6
FAN (mg/L)	BeD	33.5 ± 0.5	53.0 ± 0.9	72.7 ± 1.0	47.7 ± 0.2
	AfD	215.4 ± 32.9	236.7 ± 5.1	272.1 ± 2.2	237.7 ± 3.9
ALK (mg/L)	BeD	3114.5 ± 48.1	3415.1 ± 84.2	2950.8 ± 29.5	3233.0 ± 47.0
	AfD	6842.1 ± 599.9	6971.5 ± 166.3	7747.4 ± 66.5	8053.0 ± 232.8

Table 3
Apparent kinetic constants (k) for biogas production.

	SBP_{ad} (mL/g)	$t_{0.95}$[b] (d)	k (d^{-1})	R^2	$R_{I/S}$	p
Control	86.6 ± 3.3	20	0.19 ± 0.01	0.96	0.7	8.71E-14
Pretreatment at 60 °C	92.9 ± 1.3	11	0.29 ± 0.01	0.98	0.7	9.96E-13
Pretreatment at 70 °C	100.1 ± 5.1	5	0.39 ± 0.04	0.89	0.7	3.84E-06
Pretreatment at 80 °C	107.7 ± 8.4	5	0.39 ± 0.03	0.91	0.7	9.85E-08
No pretreatment[a]			0.33 ± 0.06	0.95	0.8	
No pretreatment[a]			0.16 ± 0.03	0.95	0.5	

[a] It refers to Raposoa et al. (2010).
[b] The time required for achieving 95% of maximum biogas yield.

the shaking water bath for the batch experiments.

Despite the limited increase in SBP_{ad}, the biogas production as a function of time changed dramatically after thermal pretreatment at low temperatures (Fig. 3), with pretreated sludge generating higher gas production rates initially than untreated sludge. This change showed that the rate-limiting step for biogas production was effectively relieved by thermal treatment, and that the rate of biogas production was accelerated by the dissolution of organic substances in sludge solids. The expression in Eq. (5) for biogas production assumes there is a proportional relationship between the degraded VS and the biogas yield (Raposoa et al., 2010).

$$G = G_m\left(1 - e^{-k\cdot t}\right) \quad (5)$$

In Eq. (5), G is the biogas production at a certain time t, G_m is the maximum biogas production, and k is the apparent kinetic constant for biogas production. Using this model with observed data, the apparent kinetic constants for biogas production can be calculated (Table 3). Statistical analyses (t-tests) showed that Eq. (5) closely predicted the experimental data. The kinetic constant doubled after short-duration thermal pretreatment at both 70 °C and 80 °C compared to the rate constant for the control (no pretreatment) samples. Due to the accelerated digestion rate, the time required for completing digestion was shortened from 20 d to 5 d after thermal pretreatment at 70 °C and at 80 °C. Considering the similar effects produced by sludge treatment at the two temperatures, yet the lower energy consumption at the lower temperature, 70 °C was identified as the optimum pretreatment temperature of those investigated.

3.4. Effect of thermal pretreatment on semi-continuous HSAD

Thermal pretreatment at 70 °C was applied to a pilot-scale digester to verify its effect on semi-continuous HSAD equipped with adequate agitation. During the whole period of anaerobic digestion, the variations of biogas production, organic removal rate and other parameters were measured, from which the average performances were statistically calculated (Table 4).

In stage 1 (initial 66 days), the system was operated steadily at a OLR of 3.4 g/(L d), which was much higher than the common range of 0.6–1.6 g/(L d) (Appels et al., 2008). Nevertheless, the pH remained stable at 7.8. The concentrations of VFAs and FAN were lower than 400 mg/L and 250 mg/L, respectively, indicating the system was operating in a satisfactory status. The specific biogas production (SBP_{ad}) of 365.2 mL/g VS added and the VS_{re} of 32.5% were very close to the previously report performance of CAD when fed a similar type of sludge at the same SRT (Li et al., 2013). The results verified that HSAD with adequate agitation can perform as well as low-solids anaerobic digestion. Hence, increasing feed TS (converting CAD to HSAD) would probably increase VBP proportionally.

After low temperature thermal pretreatment (stage 2), the average SBP_{ad} of the digester increased by 11.7% compared to that in stage 1, but the specific biogas production per unit VS removed (SBP_{re}) was almost identical to that in stage 1. Thus, the increment in biogas yield was mainly attributed to the degradation of additional organic substances released as the result of thermal treatment. However, the increase in biogas yield was limited, as had been exhibited previously in the batch experiments. Thus, even if the dissolved organic substances during thermal pretreatment had not undergone the treatment, most of them could have been utilized by anaerobic microorganism in a long digestion time. The concentrations of TN, TAN and FAN in the supernatant increased slightly, but remained lower than the thresholds that inhibit anaerobic digestion. The ratio of VFAs to TA was also very low indicating that the digester maintained a stable status. Use of thermally treated sludge did not change the acid–base balance in the digester or increase the VFAs concentration significantly. Thus, thermal pretreatment had almost no negative influence on HSAD, although the positive effect on accumulative biogas production was

Table 4
Effect of thermal pretreatment on physical and chemical parameters in high-solids sludge anaerobic digestion.

	Stage 1	Stage 2	Stage 3
Time (date)	1–66	67–136	137–175
TS (%)	15.0 ± 0.6	14.9 ± 0.8	15.4 ± 0.5
VS/TS (%)	50.4 ± 0.5	53.0 ± 1.8	54.5 ± 0.8
SRT (days)	22.0	22.0	15.0
OLR [g/(L d)]	3.4 ± 0.1	3.6 ± 0.2	5.6 ± 0.2
pH	7.8 ± 0.1	7.9 ± 0.1	7.9 ± 0.0
TA (mg/L)	6996.0 ± 248.2	8671.5 ± 803.5	10,499 ± 281.8
TN (mg/L)	2109.7 ± 266.2	2154.6 ± 485.2	2832.3 ± 707.6
TAN (mg/L)	1802.6 ± 194.3	1945.6 ± 387.9	2365.8 ± 605.7
FAN (mg/L)	123.8 ± 21.4	153.9 ± 41.4	211.2 ± 62.2
VFAs (mg/L)	184.7 ± 48.9	208.4 ± 78.5	447.0 ± 180.1
SCOD (mg/L)	6467.6 ± 1419.6	5188.2 ± 1078.7	8886.5 ± 2540.4
VFA/TA	0.03 ± 0.01	0.02 ± 0.01	0.04 ± 0.02
VS_{re} (%)	32.5 ± 2.5	36.0 ± 3.3	34.5 ± 2.3
SBP_{ad} (ml/g VS added)	332.5 ± 26.7	407.8 ± 20.5	382.7 ± 17.9
SBP_{re} (ml/g VS removed)	1026.9 ± 105.7	1139.6 ± 121.2	1114.1 ± 88.8
VBP [L/(L d)]	1.3 ± 0.1	1.5 ± 0.1	2.1 ± 0.2
CH_4/Biogas (vol.%)	54.0 ± 0.6	57.0 ± 3.9	66.6 ± 1.5

Note: Untreated sludge was digested in Stage 1, and thermally treated sludge was added in Stages 2 and 3; "±" represented the average value and the standard deviation.

limited.

However, as described in the batch experiments, some changes indeed occurred due to thermal pretreatment. In the first half of one digestion cycle, the biogas production rate (biogas yield per hour and per gram VS added) increased obviously after thermally treated sludge was fed into the digester (i.e., in stage 2) even when the SRT remained at 22 days as in stage 1, when untreated sludge was being digested (Fig. 4). It indicated that the digestion time could be shortened because of pretreatment. In fact, when the SRT was shortened to 15 days, the digester exhibited a similar performance as it did with an SRT of 22 days (Table 4). Moreover, the variation in biogas production rate in one cycle was also close to that at SRT 22 days (Fig. 4). According to Eq. (3), the average digestion degree of the thermally treated sludge was 89.8% when the SRT was 15 days (V_m and V_d were measured to be 1378 ± 239 mL and 258 ± 18 mL, respectively, using the method described in Section 2.4); this was only a little lower than the 92.4% obtained when the SRT was 22 days (V_m and V_d were 1216 ± 239 mL and 171 ± 18 mL, respectively). Based on these comparisons, low-temperature thermal pretreatment effectively accelerated the hydrolysis of substrate and shortened the digestion time from 22 days to 15 days.

On the other hand, the concentrations of TN, TAN and FAN in the supernatant all increased significantly due to the shortened digestion time (Table 4); however, they remained lower than the thresholds that cause inhibition. The average concentration of VFAs exceeded 440 mg/L, and was related to a risk of slight inhibition (Duan et al., 2012). However, a high concentration of VFAs only occurred at the beginning of stage 3, and then it decreased gradually to below 400 mg/L. The initial increase in VFAs concentration was mainly attributed to the sharp variation in OLR. The accumulated VFAs correspondingly decreased pH to 7.8–7.9. Compared with that at SRT 22 days, the biogas production rate also decreased at the initial stage of one cycle due to the shock loading (Fig. 4). Nevertheless, the system rapidly stabilized when the anaerobic microorganisms acclimated to the high-OLR environment.

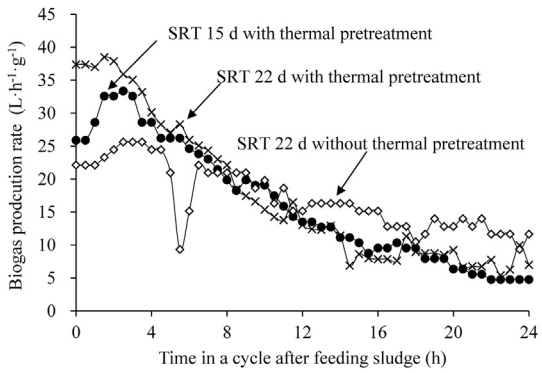

Fig. 4. Variation in biogas production rate in one digestion cycle due to thermal pretreatment.

Due to the shortened SRT, the VBP of the digester increased further to 2.1 L/(L d), which was almost 4–10 times that achievable with CAD (Appels et al., 2008). Moreover, the methane content of the biogas also increased to 66%, possibly owing to the shift of metabolic pathways under high OLR (Motte et al., 2013). Both of these variations enhance bioenergy recovery from sewage sludge. On the other hand, low-temperature thermal pretreatment increases the energy consumption of the overall anaerobic digestion process compared to digestion without thermal treatment, but the increment is much less than that resulting from high-temperature thermal pretreatment. Moreover, the increased energy consumption can be partly offset by the increased biogas production and energy savings due to shortened digestion time. For a given sludge feed rate, when the SRT is shorted from 22 to 15 days, the volume of the digester can be reduced by 32%, and correspondingly the heat dissipating surface can be reduced by about 23% (assuming a digester height is twice the diameter). In addition, results from the present study show that the stirring energy consumption was reduced due to the effect of thermal treatment in reducing sludge viscosity. On the whole, the energy consumption was greater for HSAD with thermal pretreatment than without, but the treatment capacity of digesters increased. The increased treatment capacity that was made possible by thermal pretreatment effectively reduced the footprint of a digester, which would enhance the feasibility of implementing HSAD in small WWTPs and in urban areas where the amount of physical space is limited.

4. Conclusions

Thermal hydrolysis at low temperatures is an effective pretreatment method by which to accelerate high-solids anaerobic

digestion of sewage sludge. During thermal sludge treatment, treatment temperature plays a more important role than treatment duration. After low temperature thermal treatment, a large amount of organic matter in sludge solids is dissolved, and sludge flowability is enhanced significantly. Within the range of temperatures and treatment durations investigated, thermal pretreatment at 70 °C for 30 min is optimal. Because of the accelerated hydrolysis and the decreased viscosity of thermally treated sludge, the accumulative biogas production can be increased more than 10%, and the digestion time can be shortened significantly. Combined with low-temperature thermal pretreatment, high-solids anaerobic digesters have much higher treatment capability and better applicability in small wastewater treatment plants than do conventional digesters.

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